

# BRACU CS Course Count Reference

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What each listed course counts as for the CS degree plan and calculator categories.

**Important note:** MAT110, PHY111, and STA201 count as GenEd Stream 2 courses. MAT120, MAT215, MAT216, and PHY112 count as School Core. CSE electives count under Program Electives, while CSE400 is Project/Thesis.

## Category Summary

| Counts as             | Meaning                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------|
| Program Core          | Required CSE/CS core courses and listed equivalents.                                    |
| Program Electives     | CSE elective options from the CSE department.                                           |
| School Core           | MAT120, MAT215, MAT216, PHY112. Not GenEd.                                              |
| GenEd Streams 1-5     | University Core / General Education stream courses. These make up the 13 GenEd courses. |
| Foundation / Remedial | 0-credit placement/foundation courses; do not count toward degree credits.              |

## Course Reference

### Program Core

Required CSE/CS program core courses and listed equivalents.

| Course Code | Course Title                          | Credits | Counts As    |
|-------------|---------------------------------------|---------|--------------|
| CSE 110     | Programming Language I                | 3       | Program Core |
| CSE 111     | Programming Language II               | 3       | Program Core |
| CSE 161     | Programming Language I Equivalent     | 3       | Program Core |
| CSE 162L    | Programming Language I Lab Equivalent | 0       | Program Core |
| CSE 220     | Data Structures                       | 3       | Program Core |
| CSE 221     | Algorithms                            | 3       | Program Core |
| CSE 230     | Discrete Mathematics                  | 3       | Program Core |
| CSE 260     | Digital Logic Design                  | 3       | Program Core |
| CSE 321     | Operating System                      | 3       | Program Core |
| CSE 330     | Numerical Methods                     | 3       | Program Core |

|         |                                   |   |              |
|---------|-----------------------------------|---|--------------|
| CSE 331 | Automata and Computability        | 3 | Program Core |
| CSE 340 | Computer Architecture             | 3 | Program Core |
| CSE 370 | Database Systems                  | 3 | Program Core |
| CSE 420 | Compiler Design                   | 3 | Program Core |
| CSE 421 | Computer Networks                 | 3 | Program Core |
| CSE 422 | Artificial Intelligence           | 3 | Program Core |
| CSE 423 | Computer Graphics                 | 3 | Program Core |
| CSE 470 | Software Engineering              | 3 | Program Core |
| ECE 103 | Programming Language I Equivalent | 3 | Program Core |
| ECE 283 | Digital Logic Design Equivalent   | 3 | Program Core |
| EEE 103 | Programming Language I Equivalent | 3 | Program Core |
| EEE 283 | Digital Logic Design Equivalent   | 3 | Program Core |
| EEE 301 | Digital Logic Design Equivalent   | 3 | Program Core |
| EEE 302 | Digital Logic Design Equivalent   | 0 | Program Core |

## Project / Thesis

Final project/thesis requirement.

| Course Code | Course Title     | Credits | Counts As        |
|-------------|------------------|---------|------------------|
| CSE 400     | Project & Thesis | 4       | Project / Thesis |

## Program Electives

CSE elective options from the CSE department. In your app these live in the Electives section.

| Course Code | Course Title                             | Credits | Counts As                       |
|-------------|------------------------------------------|---------|---------------------------------|
| CSE 250     | Circuits and Electronics                 | 3       | Program Elective (CSE elective) |
| CSE 251     | Electronic Devices and Circuits          | 3       | Program Elective (CSE elective) |
| CSE 310     | Object Oriented Programming              | 3       | Program Elective (CSE elective) |
| CSE 320     | Data Communication                       | 3       | Program Elective (CSE elective) |
| CSE 341     | Microprocessor                           | 3       | Program Elective (CSE elective) |
| CSE 342     | Computer Systems Engineering             | 3       | Program Elective (CSE elective) |
| CSE 350     | Digital Electronics and Pulse Techniques | 3       | Program Elective (CSE elective) |
| CSE 360     | Computer Interface                       | 3       | Program Elective (CSE elective) |
| CSE 390     | Technical Communication                  | 3       | Program Elective (CSE elective) |

|         |                                               |   |                                 |
|---------|-----------------------------------------------|---|---------------------------------|
| CSE 391 | Programming for the Internet                  | 3 | Program Elective (CSE elective) |
| CSE 392 | Signals and Systems                           | 3 | Program Elective (CSE elective) |
| CSE 410 | Advance Programming In UNIX                   | 3 | Program Elective (CSE elective) |
| CSE 419 | Programming Languages                         | 3 | Program Elective (CSE elective) |
| CSE 424 | Pattern Recognition                           | 3 | Program Elective (CSE elective) |
| CSE 425 | Neural Networks                               | 3 | Program Elective (CSE elective) |
| CSE 426 | Basic Graph Theory                            | 3 | Program Elective (CSE elective) |
| CSE 427 | Machine Learning                              | 3 | Program Elective (CSE elective) |
| CSE 428 | Image Processing                              | 3 | Program Elective (CSE elective) |
| CSE 429 | Basic Multimedia Theory                       | 3 | Program Elective (CSE elective) |
| CSE 430 | Digital Signal Processing                     | 3 | Program Elective (CSE elective) |
| CSE 431 | Natural Language Processing                   | 3 | Program Elective (CSE elective) |
| CSE 432 | Speech Recognition and Synthesis              | 3 | Program Elective (CSE elective) |
| CSE 460 | VLSI Design                                   | 3 | Program Elective (CSE elective) |
| CSE 461 | Digital System Design                         | 3 | Program Elective (CSE elective) |
| CSE 462 | Fault Tolerant Systems                        | 3 | Program Elective (CSE elective) |
| CSE 471 | System Analysis and Design                    | 3 | Program Elective (CSE elective) |
| CSE 472 | Human Computer Interface                      | 3 | Program Elective (CSE elective) |
| CSE 473 | Decision Support System                       | 3 | Program Elective (CSE elective) |
| CSE 474 | Simulation and Modeling                       | 3 | Program Elective (CSE elective) |
| CSE 489 | Special Topics                                | 3 | Program Elective (CSE elective) |
| CSE 490 | WAN Routing and Technologies / Special Topics | 3 | Program Elective (CSE elective) |
| CSE 491 | Independent Study                             | 3 | Program Elective (CSE elective) |

## School Core

Required school core math/physics courses. These are not GenEd stream courses.

| Course Code | Course Title                                            | Credits | Counts As   |
|-------------|---------------------------------------------------------|---------|-------------|
| MAT 120     | MATH II: Integral Calculus and Differential Equations   | 3       | School Core |
| MAT 215     | MATH III: Complex Variables and Laplace Transformations | 3       | School Core |
| MAT 216     | MATH IV: Linear Algebra and Fourier Analysis            | 3       | School Core |

|         |                          |   |             |
|---------|--------------------------|---|-------------|
| PHY 112 | Principles of Physics II | 3 | School Core |
|---------|--------------------------|---|-------------|

### Foundation / Remedial

Zero-credit placement/foundation courses. They do not count toward the 124-credit total.

| Course Code | Course Title                   | Credits | Counts As                                                         |
|-------------|--------------------------------|---------|-------------------------------------------------------------------|
| ENG 091     | Foundation Course              | 0       | Foundation/remedial (0 credit; does not count toward 124 credits) |
| MAT 092     | Remedial Course in Mathematics | 0       | Foundation/remedial (0 credit; does not count toward 124 credits) |

### GenEd Stream 1: Writing Comprehension

University Core / General Education stream courses.

| Course Code | Course Title                             | Credits | Counts As                             |
|-------------|------------------------------------------|---------|---------------------------------------|
| ENG 101     | English Fundamentals                     | 3       | GenEd Stream 1: Writing Comprehension |
| ENG 102     | English Composition I                    | 3       | GenEd Stream 1: Writing Comprehension |
| ENG 103     | Advanced Writing Skills and Presentation | 3       | GenEd Stream 1: Writing Comprehension |

### GenEd Stream 2: Math and Natural Sciences

University Core / General Education stream courses.

| Course Code | Course Title                                           | Credits | Counts As                                 |
|-------------|--------------------------------------------------------|---------|-------------------------------------------|
| BIO 101     | Introduction to Biology                                | 3       | GenEd Stream 2: Math and Natural Sciences |
| CHE 101     | Introduction to Chemistry                              | 3       | GenEd Stream 2: Math and Natural Sciences |
| CSE 101     | Introduction to Computer Science                       | 3       | GenEd Stream 2: Math and Natural Sciences |
| ENV 103     | Elements of Environmental Science                      | 3       | GenEd Stream 2: Math and Natural Sciences |
| MAT 101     | Fundamentals of Mathematics                            | 3       | GenEd Stream 2: Math and Natural Sciences |
| MAT 110     | MATH I: Differential Calculus and Co-ordinate Geometry | 3       | GenEd Stream 2: Math and Natural Sciences |
| PHY 101     | Introduction to Physics                                | 3       | GenEd Stream 2: Math and Natural Sciences |

|         |                                          |   |                                           |
|---------|------------------------------------------|---|-------------------------------------------|
| PHY 111 | Principles of Physics I                  | 3 | GenEd Stream 2: Math and Natural Sciences |
| STA 101 | Introduction to Statistics               | 3 | GenEd Stream 2: Math and Natural Sciences |
| STA 201 | Elements of Statistics and Probabilities | 3 | GenEd Stream 2: Math and Natural Sciences |

### GenEd Stream 3: Arts and Humanities

University Core / General Education stream courses.

| Course Code | Course Title                                       | Credits | Counts As                           |
|-------------|----------------------------------------------------|---------|-------------------------------------|
| BNG 103     | Bangla Language and Literature                     | 3       | GenEd Stream 3: Arts and Humanities |
| ENG 110     | English for Life                                   | 3       | GenEd Stream 3: Arts and Humanities |
| ENG 113     | Introduction to English Poetry                     | 3       | GenEd Stream 3: Arts and Humanities |
| ENG 114     | Introduction to English Drama                      | 3       | GenEd Stream 3: Arts and Humanities |
| ENG 115     | Introduction English Prose                         | 3       | GenEd Stream 3: Arts and Humanities |
| ENG 333     | Globalization and the Media                        | 3       | GenEd Stream 3: Arts and Humanities |
| HST 102     | The Modern World                                   | 3       | GenEd Stream 3: Arts and Humanities |
| HST 103     | History of Bangladesh                              | 3       | GenEd Stream 3: Arts and Humanities |
| HST 104     | Global History Lab - A History of World since 1300 | 3       | GenEd Stream 3: Arts and Humanities |
| HUM 101     | World Civilization & Culture                       | 3       | GenEd Stream 3: Arts and Humanities |
| HUM 102     | Introduction to Philosophy                         | 3       | GenEd Stream 3: Arts and Humanities |
| HUM 103     | Ethics and Culture                                 | 3       | GenEd Stream 3: Arts and Humanities |
| HUM 207     | Narratives of Truth and Lies                       | 3       | GenEd Stream 3: Arts and Humanities |
| HUM 210     | Streaming Media Cultures                           | 3       | GenEd Stream 3: Arts and Humanities |

|         |                 |   |                                     |
|---------|-----------------|---|-------------------------------------|
| HUM 301 | In the Archives | 3 | GenEd Stream 3: Arts and Humanities |
|---------|-----------------|---|-------------------------------------|

## GenEd Stream 4: Social Sciences

University Core / General Education stream courses.

| Course Code | Course Title                                       | Credits | Counts As                       |
|-------------|----------------------------------------------------|---------|---------------------------------|
| ANT 101     | Introduction to Anthropology                       | 3       | GenEd Stream 4: Social Sciences |
| ANT 342     | Body and Society                                   | 3       | GenEd Stream 4: Social Sciences |
| ANT 351     | Gender & Development                               | 3       | GenEd Stream 4: Social Sciences |
| BUS 102     | Business - Basics, Ethics and Environment          | 3       | GenEd Stream 4: Social Sciences |
| BUS 201     | Business and Human Communication                   | 3       | GenEd Stream 4: Social Sciences |
| BUS 333     | Social Entrepreneurship Practicum                  | 3       | GenEd Stream 4: Social Sciences |
| BUS 334     | Social Intrapreneurship Practicum - Leading Change | 3       | GenEd Stream 4: Social Sciences |
| BUS 335     | Sustainable Development and Social Enterprise      | 3       | GenEd Stream 4: Social Sciences |
| DEV 101     | Bangladesh Studies                                 | 3       | GenEd Stream 4: Social Sciences |
| DEV 104     | Foundations of International Development           | 3       | GenEd Stream 4: Social Sciences |
| DEV 201     | Health, Culture and Development                    | 3       | GenEd Stream 4: Social Sciences |
| ECO 101     | Introduction to Microeconomics                     | 3       | GenEd Stream 4: Social Sciences |
| ECO 102     | Introduction to Macroeconomics                     | 3       | GenEd Stream 4: Social Sciences |
| ECO 105     | Introduction to Economics                          | 3       | GenEd Stream 4: Social Sciences |
| EMB 101     | Emergence of Bangladesh                            | 3       | GenEd Stream 4: Social Sciences |
| POL 101     | Introduction to Political Science                  | 3       | GenEd Stream 4: Social Sciences |
| POL 102     | Comparative Governance                             | 3       | GenEd Stream 4: Social Sciences |
| POL 103     | International Relations and Global Politics        | 3       | GenEd Stream 4: Social Sciences |
| POL 201     | Introduction to Civic Engagement                   | 3       | GenEd Stream 4: Social Sciences |
| POL 202     | Foundations in Public Policy                       | 3       | GenEd Stream 4: Social Sciences |
| POL 203     | Political Theory                                   | 3       | GenEd Stream 4: Social Sciences |
| POL 210     | Youth and Global Movement                          | 3       | GenEd Stream 4: Social Sciences |
| PSY 101     | Introduction to Psychology                         | 3       | GenEd Stream 4: Social Sciences |
| PSY 102     | Understanding the Human Minds                      | 3       | GenEd Stream 4: Social Sciences |
| SOC 101     | Introduction to Sociology                          | 3       | GenEd Stream 4: Social Sciences |

|                   |                   |   |                                 |
|-------------------|-------------------|---|---------------------------------|
| SOC 201 / ANT 202 | Social Inequality | 3 | GenEd Stream 4: Social Sciences |
|-------------------|-------------------|---|---------------------------------|

## GenEd Stream 5: Communities, Seeking Transformation

University Core / General Education stream courses.

| Course Code | Course Title                                                        | Credits | Counts As                                           |
|-------------|---------------------------------------------------------------------|---------|-----------------------------------------------------|
| BU 201      | Community Engagement and Action                                     | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 201     | Luxury and Oppression in a Globalized World (of Perfumes)           | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 204     | Communication for Social Change                                     | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 301     | For the Love of Food                                                | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 302     | The Pursuit of Wellbeing                                            | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 303     | Law for Life, Peace and Justice                                     | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 304     | Documentary Film: Theory and Practice                               | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 305     | Borders and Beyond: Past and Future                                 | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 306     | Ethical Leadership                                                  | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 307     | Art, Community and the Future                                       | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 308     | Social Dimensions of Faith and Development                          | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 309     | Global Citizenship                                                  | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 310     | Social Cohesion and Peace Building                                  | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 314     | Visual Storytelling: Understanding Society through Documentary Film | 3       | GenEd Stream 5: Communities, Seeking Transformation |
| CST 333     | Personal Finance for Sustainable Economic Wellbeing                 | 3       | GenEd Stream 5: Communities, Seeking Transformation |